

## Platform Overload Components

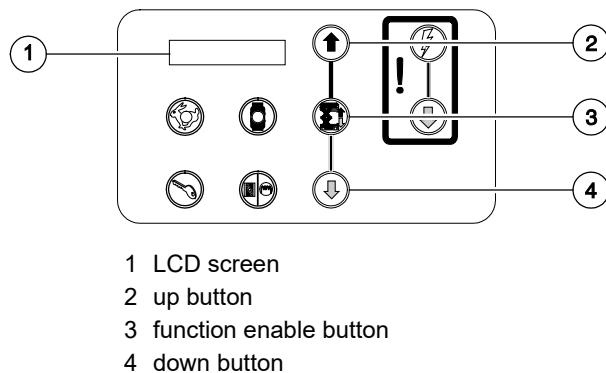
### 14-2

#### Platform Overload Recovery Message (software version B3 and later)

If the ground controls LCD screen displays OVRLD RECOVERY, the emergency lowering system has been used while the platform was overloaded.

Note: This message shall be cleared by a person trained and qualified on the troubleshooting and repair of this machine.

Note: Use the following illustration to identify the description of each control button used in this procedure.



- 1 Turn the key switch to the ground controls position and pull out the red Emergency Stop button to the on position at the platform controls.
- 2 Press and hold both the up and down buttons. Pull out the red Emergency Stop button to the on position at the ground controls.
- Result: TUNE SPEEDS is showing in the LCD screen.

- 3 Use the down button to scroll to select options.
- 4 Press the function enable button.
- Result: DESCENT DELAY is showing in the LCD screen.
- 5 Use the down button to scroll to overload.
- 6 Press and hold the function enable button for 5 seconds.
- Result: A screen will appear asking for the overload recovery passcode.
- 7 Press the buttons in the following sequence: **(down) (down) (up) (enable)**
- Result: The LCD screen displays CLEARED and returns to the OVERLOAD SCREEN.
- 8 Use the down button to scroll to RETURN TO MAIN MENU.
- 9 Press the function enable button.
- Result: SELECT OPTIONS is showing in the LCD screen.
- 10 Push in the red Emergency stop button at the ground controls.

## Fault Codes



### Observe and Obey:

- ☑ Troubleshooting and repair procedures shall be completed by a person trained and qualified on the repair of this machine
- ☑ Immediately tag and remove from service a damaged or malfunctioning machine.
- ☑ Repair any machine damage or malfunction before operating the machine.
- ☑ Unless otherwise specified, perform each procedure with the machine in the following configuration:
  - Machine parked on a firm, level surface
  - Key switch in the off position with the key removed
  - The red Emergency Stop button in the off position at both the ground and platform controls
  - Wheels chocked
  - All external AC power supply disconnected from the machine
  - Platform in the stowed position

### Before Troubleshooting:

- ☑ Read, understand and obey the safety rules and operating instructions in the appropriate operator's manual on your machine.
- ☑ Be sure that all necessary tools and test equipment are available and ready for use.
- ☑ Read each appropriate fault code thoroughly. Attempting short cuts may produce hazardous conditions.
- ☑ Be aware of the following hazards and follow generally accepted safe workshop practices.

#### **⚠ DANGER**

Crushing hazard. When testing or replacing any hydraulic component, always support the structure and secure it from movement

#### **⚠ WARNING**

Electrocution/burn hazard. Contact with electrically charged circuits could result in death or serious injury. Remove all rings, watches and other jewelry.

#### **⚠ WARNING**

Bodily injury hazard. Spraying hydraulic oil can penetrate and burn skin. Loosen hydraulic connections very slowly to allow the oil pressure to dissipate gradually. Do not allow oil to squirt or spray.

- ⦿ Indicates that a specific result is expected after performing a series of steps.
- ⊗ Indicates that an incorrect result has occurred after performing a series of steps.

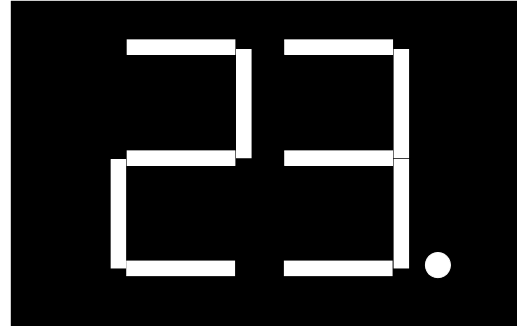
Note: Two persons will be required to safely perform some troubleshooting procedures.

## Fault Codes

### About This Section

When a malfunction is discovered, the fault code charts in this section will help a service professional pinpoint the cause of the problem. To use this section, basic hand tools and certain pieces of test equipment are required — voltmeter, ohmmeter, pressure gauges.

### LED Diagnostic Readout



The diagnostic readout displays numerical codes that provide information about the machine operating status and about malfunctions. The dot to the right of the numbers remain on when a fault code is displayed.

The codes listed in the Fault Code Chart describe malfunctions and can aid in troubleshooting the machine by pinpointing the area or component affected.

### Platform LED

Green - System ready, normal operation.

Red - A fault condition has occurred.

## Fault Codes

| ID | System Ready               | LED | Condition                                                                  | Result                                  | Solution                                                                        |
|----|----------------------------|-----|----------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------|
| 1  | INTERNAL ECU FAULT         | Red | Internal ECM error.                                                        | System shutdown.                        | Replace ECM.                                                                    |
| 2  | PLATFORM ECU FAULT         | Red | Platform/ECM communication error.                                          | System shutdown.                        | Troubleshoot control cable OR troubleshoot platform controls.                   |
| 20 | CHASSIS START SW FAULT     | Red | Engine start button fault at ground controls.                              | Engine will not start.                  | Replace ECM.                                                                    |
| 21 | CHASSIS CHOKE SW FAULT     | Red | Starting aid button fault at ground controls.                              | Starting aid disabled.                  | Replace ECM.                                                                    |
| 22 | CHASSIS UP SW FAULT        | Red | Up switch fault at ground controls.                                        | Platform up function inhibited.         | Replace ECM.                                                                    |
| 23 | CHASSIS LIFT SW FAULT      | Red | Platform up/down enable button fault at ground controls.                   | Platform up/down functions inhibited.   | Replace ECM.                                                                    |
| 24 | DOWN SW FAULT              | Red | Down switch fault at ground controls.                                      | Platform down function inhibited.       | Replace ECM.                                                                    |
| 25 | LEFT TURN SW FAULT         | Red | Left turn switch fault.                                                    | Malfunctioning steer left microswitch.  | Troubleshoot steer left microswitch.                                            |
| 26 | RIGHT TURN SW FAULT        | Red | Right turn switch fault.                                                   | Malfunctioning steer right microswitch. | Troubleshoot steer right microswitch.                                           |
| 27 | DRIVE ENABLE SW FLT        | Red | Function enable switch on joystick is activated when machine is turned on. | Machine functions disabled.             | Release function enable switch on joystick before power up OR replace joystick. |
| 28 | OFF NEUTRAL DRIVE JOYSTICK | Red | Drive joystick off neutral when machine is turned on.                      | Machine functions disabled.             | Release joystick before power up OR replace joystick.                           |
| 29 | PLATFORM LIFT SW FAULT     | Red | Lift enable button fault at the platform controls.                         | Platform up/down functions inhibited.   | Troubleshoot button OR replace printed circuit board at platform controls.      |
| 30 | OFF NEUTRAL LIFT JOYSTICK  | Red | Up/down switch off neutral.                                                | Platform up/down functions inhibited.   | Replace up/down switch at platform controls.                                    |
| 31 | PLATFORM CHOKE SW FAULT    | Red | Starting aid fault at the platform controls.                               | Starting aid disabled.                  | Troubleshoot button OR replace printed circuit board at platform controls.      |

## Fault Codes

| ID   | System Ready              | LED | Condition                                                                                                                                                                | Result                                      | Solution                                                                                                                   |
|------|---------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 32   | PLATFORM START SW FAULT   | Red | Engine start button fault at platform controls.                                                                                                                          | Engine will not start.                      | Troubleshoot button OR replace printed circuit board at platform controls.                                                 |
| 33   | LEFT FRONT OUTRIG SW FLT  | Red | Left front outrigger enable button fault at platform controls.                                                                                                           | Outriggers disabled.                        | Troubleshoot button OR replace printed circuit board at platform controls.                                                 |
| 34   | RIGHT FRONT OUTRIG SW FLT | Red | Right front outrigger enable button fault at platform controls.                                                                                                          | Outriggers disabled.                        | Troubleshoot button OR replace printed circuit board at platform controls.                                                 |
| 35   | LEFT REAR OUTRIG SW FLT   | Red | Left rear outrigger enable button fault at platform controls.                                                                                                            | Outriggers disabled.                        | Troubleshoot button OR replace printed circuit board at platform controls.                                                 |
| 36   | RIGHT REAR OUTRIG SW FLT  | Red | Right rear outrigger enable button fault at platform controls.                                                                                                           | Outriggers disabled.                        | Troubleshoot button OR replace printed circuit board at platform controls.                                                 |
| 37   | AUTO LEVEL SWITCH FAULT   | Red | Outrigger auto level enable button fault at platform controls.                                                                                                           | Outriggers disabled.                        | Troubleshoot button OR replace printed circuit board at platform controls.                                                 |
| 3812 | OSCILLATE LIMIT SWITCHES  | Red | Value to high. Voltage above 3000 mV.                                                                                                                                    | Elevated drive and platform up disabled.    | Troubleshoot oscillate limit switches.                                                                                     |
| 3815 | OSCILLATE LIMIT SWITCHES  | Red | Value to low. Voltage below 1000 mV.                                                                                                                                     | Elevated drive and platform up disabled.    | Troubleshoot oscillate limit switches.                                                                                     |
| 3819 | OSCILLATE LIMIT SWITCHES  | Red | Voltage out of range.                                                                                                                                                    | Elevated drive and platform up disabled.    | Troubleshoot oscillate limit switches.                                                                                     |
| 3821 | OSCILLATE LIMIT SWITCHES  | Red | Voltage inconsistent.                                                                                                                                                    | Elevated drive and platform up disabled.    | Troubleshoot oscillate limit switches.                                                                                     |
| 4021 | OSCILLATE OUTPUT          | Red | Both oscillate valve outputs are on at the same time while elevated.<br><br>Flashing or Motion Beacon option is set to <b>on</b> and it is not installed on the machine. | Elevated drive and platform up disabled.    | Troubleshoot oscillate limit switches or replace ECU.<br><br>Set the option to <b>off</b> if not installed on the machine. |
| 4022 | OSCILLATE OUTPUT          | Red | Oscillate valve output is active for more than 4 seconds.                                                                                                                | Elevated drive and platform up disabled.    | Troubleshoot oscillate limit switches or replace ECU.                                                                      |
| 49   | DRIVE COIL 1 FAULT        | Red | Drive coil 1 (Y1) fault.                                                                                                                                                 | High speed drive function will not operate. | Troubleshoot coil OR wiring.                                                                                               |

## Fault Codes

| ID | System Ready          | LED | Condition                      | Result                                      | Solution                                                                                                           |
|----|-----------------------|-----|--------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 50 | DRIVE COIL 2 FAULT    | Red | Drive coil 2 (Y1A) fault.      | High speed drive function will not operate. | Troubleshoot coil OR wiring.                                                                                       |
| 51 | DRIVE COIL 3 FAULT    | Red | Drive coil 3 (Y1B) fault.      | High speed drive function will not operate. | Troubleshoot coil OR wiring.                                                                                       |
| 52 | FUNC PROP COIL FAULT  | Red | Proportional coil (Y9) fault.  | Lift and outrigger functions are disabled.  | Troubleshoot coil OR wiring.                                                                                       |
| 54 | UP COIL FAULT         | Red | Up coil (Y8) fault.            | Platform up function inhibited.             | Troubleshoot coil OR wiring.                                                                                       |
| 55 | DOWN COIL FAULT       | Red | Down coil (Y7) fault.          | Platform down function inhibited.           | Troubleshoot coil OR wiring.                                                                                       |
| 56 | RIGHT TURN COIL FAULT | Red | Right turn coil (Y3) fault.    | Steer right function inhibited.             | Troubleshoot coil OR wiring.                                                                                       |
| 57 | LEFT TURN COIL FAULT  | Red | Left turn coil (Y4) fault.     | Steer left function inhibited.              | Troubleshoot coil OR wiring.                                                                                       |
| 58 | BRAKE COIL FAULT      | Red | Brake release coil (Y2) fault. | Brakes will not release.                    | Troubleshoot coil OR wiring.                                                                                       |
| 60 | FORWARD 1 COIL FAULT  | Red | Forward 1 coil (Y6) fault.     | Drives slow or not at all when elevated.    | Troubleshoot coil OR wiring.                                                                                       |
| 61 | REVERSE 1 COIL FAULT  | Red | Reverse 1 coil (Y5) fault.     | Drives slow or not at all when elevated.    | Troubleshoot coil OR wiring.                                                                                       |
| 62 | FORWARD 2 COIL FAULT  | Red | Forward 2 coil (Y6A) fault.    | Drives slow or not at all when elevated.    | Troubleshoot coil OR wiring.                                                                                       |
| 63 | REVERSE 2 COIL FAULT  | Red | Reverse 2 coil (Y5A) fault.    | Drives slow or not at all when elevated.    | Troubleshoot coil OR wiring.                                                                                       |
| 64 | OSCILLATE LEFT COIL   | Red | Value to low or Value to high. | Elevated drive and platform up disabled.    | Troubleshoot oscillate left valve coil (Y93) or wiring.                                                            |
| 65 | OSCILLATE RIGHT COIL  | Red | Value to low or Value to high. | Elevated drive and platform up disabled.    | Troubleshoot oscillate right valve coil (Y94) or wiring.                                                           |
| 66 | LOW OIL PRESSURE      | Red | Low oil pressure.              | Engine stops.                               | Check the engine oil level OR check wiring from the oil pressure switch to ECM OR replace the oil pressure switch. |

## Fault Codes

| ID  | System Ready              | LED | Condition                                        | Result                                                                         | Solution                                                                                                                                                                                                                                                                                                          |
|-----|---------------------------|-----|--------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 67  | HIGH COOLANT TEMPERATURE  | Red | High coolant temperature.                        | High engine temperature OR defective engine coolant or oil temperature switch. | Gasoline/LPG models: Check the engine radiator coolant level OR check the wiring from the water temperature switch to ECM OR replace the water temperature switch.<br>Diesel models: Check the engine oil level OR check the wiring from the oil temperature switch to ECM OR replace the oil temperature switch. |
| 68  | LOW ECU VOLTAGE           | Red | Low ECM voltage.                                 | System shutdown.                                                               | Charge battery.                                                                                                                                                                                                                                                                                                   |
| 69  | LOW ENGINE RPM            | Red | Low RPM.                                         | Engine idle RPM too low.                                                       | Contact Genie Product Support.                                                                                                                                                                                                                                                                                    |
| 70  | HIGH ENGINE RPM           | Red | High RPM.                                        | Engine RPM too high.                                                           | Contact Genie Product Support.                                                                                                                                                                                                                                                                                    |
| 80  | LEFT FRONT OTRG COIL FLT  | Red | Left front outrigger coil (Y35) fault.           | Left front outrigger function inhibited.                                       | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 81  | LEFT REAR OTRG COIL FLT   | Red | Left rear outrigger coil (Y33) fault.            | Left front outrigger function inhibited.                                       | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 82  | RIGHT FRONT OTRG COIL FLT | Red | Right front outrigger coil (Y36) fault.          | Right front outrigger function inhibited.                                      | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 83  | RIGHT REAR OTRG COIL FLT  | Red | Right rear outrigger coil (Y34) fault.           | Right rear outrigger function inhibited.                                       | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 84  | OUTRIGGER EXT COIL FLT    | Red | Outrigger extend coil (Y40) fault.               | Outrigger extend function disabled.                                            | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 85  | OUTRIGGER RET COIL FLT    | Red | Outrigger retract coil (Y39) fault.              | Outrigger retract function disabled.                                           | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 86  | OUTRIGGER SLOW COIL FLT   | Red | Outrigger slowdown coil (Y44) fault.             | Outrigger slow extend function disabled.                                       | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 90  | 2 SPEED COIL FAULT        | Red | 2 speed coil (Y1) fault                          | High torque drive function disabled.                                           | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 92  | DRIVE FWD PROP COIL FAULT | Red | Drive pump forward proportional coil (Y51) fault | Drive forward function disabled.                                               | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 93  | DRIVE REV PROP COIL FAULT | Red | Drive pump reverse proportional coil (Y51) fault | Drive reverse function disabled.                                               | Troubleshoot coil OR wiring.                                                                                                                                                                                                                                                                                      |
| 94  | MACHINE TYPE FAULT        | Red | Wrong machine type selected                      | Machine functions disabled.                                                    | Correct selection.                                                                                                                                                                                                                                                                                                |
| 102 | DOWN LS CROSS-CHECK FLT   | Red | Down limit switch fault                          | All functions inhibited except platform down.                                  | Troubleshoot down limit switches and wiring.                                                                                                                                                                                                                                                                      |